EncroChat – EIO Data Review

Luke Shrimpton

Date 20/10/21

Version number 0.1

I Luke William Shrimpton DECLARE THAT:

1. I understand that my duty is to help the court to achieve the overriding objective by giving independent assistance by way of objective, unbiased opinion on matters within my expertise, both in preparing reports and giving oral evidence. I understand that this duty overrides any obligation to the party by whom I am engaged or the person who has paid or is liable to pay me. I confirm that I have complied with and will continue to comply with that duty.
2. I confirm that I have not entered into any arrangement where the amount or payment of my fees is in any way dependent on the outcome of the case.
3. I know of no conflict of interest of any kind, other than any which I have disclosed in my report.
4. I do not consider that any interest which I have disclosed affects my suitability as an expert witness on any issues on which I have given evidence.
5. I will advise the party by whom I am instructed if, between the date of my report and the trial, there is any change in circumstances which affect my answers to points 3 and 4 above.
6. I have shown the sources of all information I have used.
7. I have exercised reasonable care and skill in order to be accurate and complete in preparing this report.
8. I have endeavoured to include in my report those matters, of which I have knowledge or of which I have been made aware, that might adversely affect the validity of my opinion. I have clearly stated any qualifications to my opinion.
9. I have not, without forming an independent view, included or excluded anything which has been suggested to me by others including my instructing lawyers.
10. I will notify those instructing me immediately and confirm in writing if for any reason my existing report requires any correction or qualification.
11. I understand that:
    1. my report will form the evidence to be given under oath or affirmation;
    2. the court may at any stage direct a discussion to take place between experts;
    3. the court may direct that, following a discussion between the experts, a statement should be prepared showing those issues which are agreed and those issues which are not agreed, together with the reasons;
    4. I may be required to attend court to be cross-examined on my report by a cross-examiner assisted by an expert.

5. I am likely to be the subject of public adverse criticism by the judge if the Court concludes that I have not taken reasonable care in trying to meet the standards set out above.

12. I have read Part 19 of the Criminal Procedure rules and I have complied with its requirements.
13. I confirm that I have read guidance contained in a booklet known as Disclosure: Experts' Evidence and Unused Material which details my role and documents my responsibilities, in relation to disclosure as an expert witness. I have followed the guidance and recognise the continuing nature of my responsibilities of revelation. In accordance with my duties of disclosure, as documented in the guidance booklet, I confirm that:
    1. I have complied with my duties to record, retain and reveal material in accordance with the Criminal Procedure and Investigations Act 1996, as amended;
    2. I have compiled an Index of all material. I will ensure that the Index is updated in the event I am provided with or generate additional material;
    3. In the event my opinion changes on any material issue, I will inform the investigating officer, as soon as reasonably practicable and give reasons.

STATEMENT OF TRUTH

I confirm that the contents of this report are true to the best of my knowledge and belief and that I make this report knowing that, if it is tendered in evidence, I would be liable to prosecution if I have wilfully stated anything which I know to be false or that I do not believe to be true.

Luke William Shrimpton

I have reviewed the exhibit B39/2020/00267/114. The data consists of folders for each handle; each folder contains one or two Excel worksheets and a further folder containing images.

To ensure I am clear as to which worksheet is being referred to in my report I refer to them using the following names:

1) **Device Worksheet**: This contains tabs including Notes, WiFi and Cell IDs.
2) **Messages Worksheet**: This contains messages, no information such as Notes, WiFi and Cell IDs are included.

1. In each case where there are two worksheets is there one with tabs that contains the device details for the handle in question along with contacts, messages, notes etc?

Yes, when there are two worksheets there is one Device Worksheet and one Messages Worksheet

2. Is there any data contained in the worksheets that is not contained in the json_data files which were provided in relation to other devices?

I have identified two bits of data that do not easily match up to data in the json_data files, they both relate to the sheet that ends with the word 'Global'.

Account ID:

This is a v4 (random) UUID. I do not know exactly where this was taken from though I believe it could be the account ID stored in the Realm database which is also a v4 UUID.

Hash:

This appears to be hex encoded data (characters 0-9 and a-f) though it also contains a small number of asterixes which are not hex characters. I do not know what this relates to. It does not appear similar to any data present in packs for other handles. It also doesn't appear to be the result or any standard hash function (e.g. MD5, SHA1, SHA256). I have reviewed the app to attempt to identify any data that would be formatted like the hash and was unable to find anything.

In the Messages tab the MD5 column is a reference for the images in the folder, it appears to be an MD5 hash (correct length and it is common to present MD5 values as hex encoded data) and is the MD5 hash for the decrypted image. For example in the workbook ‘amazon_world2.xlsx ’in the tab gb_Messages an image is sent from buckgrove to amazon-world with the MD5 hash of 649b48f4715f486d7be584293fecfb56 at the time Tue May 12 12:05:56 GMT+01:00 2020. An image with the same MD5 hash can be found in the pack for IMEI 358098093386896 (buckgrove), image ID d1be5e5d-9a19-4c1f-a466-c140cd23c06f with the dateCreated of Tue May 12 12:05:56 GMT+01:00 2020. The image file names are truncated hash values where the last five characters of the hash value have been removed. I don’t know why this was done though it is clear there is no extra information used to create the hash values.

While there are sheets for Comments and Calendar which is not included in the json_data files they contain no data. Similarly on the ‘Global ’tab there are column labelled Tags and Identification which are also empty.

3. Could my script be rewritten to produce a workbook similar to the ones being reviewed?

Yes.

4. Is there anything in the data which suggests these files were used to create the json_data files?

No. the json_data files contain more information than these workbooks for example message id, packet id, dateRead. These workbooks could not be used to generate the json_data files.

5. Do you still believe the data obtained from the handsets was stored in the json_data files which were then used to create the CSV files?

Yes. The json_data files contain more information than the workbooks. The workbooks would have to be generated from the json_data files.

6. Is it possible to say from the data whether the messages in the Messages Workbook was taken from a different source as the messages in the Device Workbook? Does looking at where messages appear in both workbooks help with this assessment?

Due to the highly processed nature of the data and the limited fields present in the data it is difficult to make a technical assessment about the source of the data.

Unlike the raw data provided to us for devices exploited in the UK it doesn't include information about the source of a message (e.g.: The IMEI of the device where the message was collected).

If a row containing a message in the Devices Workbook exactly matches a row in the Messages Workbook I can't make a technical assessment as to whether it was collected from the same device or a different device as the message could have travelled between devices within a second.

If the same message (To, From and body/MD5) was collected with different timestamps it would indicate the message was collected from different devices or the message was sent twice from the same device. With the data available I am unable to make a technical assessment as to what happened.

I can't comment on the workbooks having different sources.

7. Could the Messages Workbook be generated in the same way 'virtual packs 'have been generated by UK Law Enforcement?

Yes, 'virtual packs 'similar to the Messages Workbook can be generated by identifying all messages sent to/from a user over all packs and combining them into a single file.

8. Is the Messages Workbook in the same format as the Device Workbook?

The Device Workbook includes tabs that contain messages. These messages tabs appear to be in the same format. They both have the same column headers: Sender, Timezone, Recipient, Burn (days), Body, MD5, Device date.

There are a number of similar features between the messages tabs such as the time formats, the fact that the burn time is missing on some data and the device date is missing on some data. These indicate the data is in the same format.

9. Does the data in any way support the hypothesis that the data was created from intercepted copies of messages where metadata was correlated with messages?

The Message sheets within the workbooks only contains information that it is possible to collect from intercept[1] (From, To, Burn, Time and Body) unlike the 'json_data 'files which contain fields like message id, read information and nicknames which can't be collected from intercept.

The data is clearly processed data, burn time is sent and stored in seconds, and the workbooks refer to burn time in days.

All the content in the workbooks (excluding the Hash) is data that appears in the original packs, which comes from the device. The data is presented in a different format which is the result of a different processing script. I am unsure what the Hash refers to but I have not seen anything like it during my experiments with EncroChat including the analysis of test intercept. The presence of this hash does not indicate the data was collected from intercept.

While reviewing the data I noticed that sometimes messages do not have a burn time or a device date.

I have considered if the lack of burn / device time provides evidence of intercept:

All messages are both sent and stored with a burn time, the lack of a burn time does not indicate the message was collected from intercept.

Messages are stored with at least a dateCreated, messages are not sent with a time, and this would have to be added as messages are collected. Associating an intercepted message with its time is a trivial and critical processing task, if a message was collected via intercept I would still expect to see a time associated to it. I don't believe the lack of time is evidence for intercept.

I believe the lack of burn time or device date is not evidence of intercept it is more likely to be an artefact of the French processing script.

There is no message specific metadata that would require a 'correlation 'tool.

[1] Body would only be available if a technique to bypass the cryptography was introduced.

10. Does the data in any way suggest it did not come from the handset?

For the data to suggest it didn't come from the device it could:

a) Contain information that could not be explained as being produced by running a different processing script over raw data like what was collected from devices in the UK. The only bit of data that can't be explained by this is the Hash as discussed in questions 2 and 9.

b) Contain artefacts that suggest the data did not come from processing the raw data like what was collected from devices in the UK. The lack of burn time and device time for some messages is such an artefact. It is easy to extract this from the raw data ('json_data 'or Realm). I don't believe the lack of this data is indicative of intercept or collection of data from somewhere other than the device as there is nowhere a message is sent/stored where it does not have its burn time associated to it and it would not be trivial to associate a time to the message.

I don't know why messages are missing burn time / device time. One possibility is they are deleted messages that were recovered using a different tool for deleted message recovery (similar to what is presented in DeletedMessages.csv). Another possibility is issues with the French processing script.

11. Does the data in any way suggest it came from the server as opposed to the handset?

See answer to 10.

12. Would I be able to produce a database that allowed me to produce a workbook containing all messages to/from a handle irrespective of the device the data was collected from?

Yes.

13. If I created a database as described in 12 would it suggest the data had been drawn from a different source or there had been a correlation of messages and metadata.

No.

14. Are the defence correct in asserting that the only data in the whitestuff pack are messages and data present in the amazonworld pack.

I have reviewed the data associated to IMEI 358098092409467 (whitestuff) it contains data that is not in ‘amazon_world.xlsx ’or ‘amazon_world2.xlsx ’for example it contains the cell ids associated to the device with IMEI 358098092409467 (whitestuff), it also contains the IMSI and SIM number (ICCID) associated with the IMEI 358098092409467 and messages to contacts other than amazon-world that do not appear in the amazon-world pack.

15. Do messages sent by whitestuff to amazonworld appear in the whitestuff pack?

I have reviewed the data associated to IMEI 358098092409467 (whitestuff) it contains messages from whitestuff to amazonworld, for example the message with id b19a6dc7-77a3-4098-bebf-57db49249405.

16. Are there any messages in the Messages Workbook for amazon-world that are to/from whitestuff but don’t appear in the whitestuff pack?

I have reviewed the data in ‘amazon_world2.xlsx ’and the data associated to IMEI 358098092409467 (whitestuff) extracted all messages to/from amazon-world and whitestuff normalised the output to the same format. It produces exactly the same results indicating they both contain the same messages to/from amazon-world and whitestuff.

17. Are there any discrepancies between the messages to/from amazon-world and whitestuff from ‘amazon_world2.xlsx ’and the data associated to the IMEI 358098092409467 (whitestuff). If so do these discrepancies indicate the data came from intercept.

There are no discrepancies.

Other Comments:

I have no knowledge of the processing script used by the French to produce the data provided.